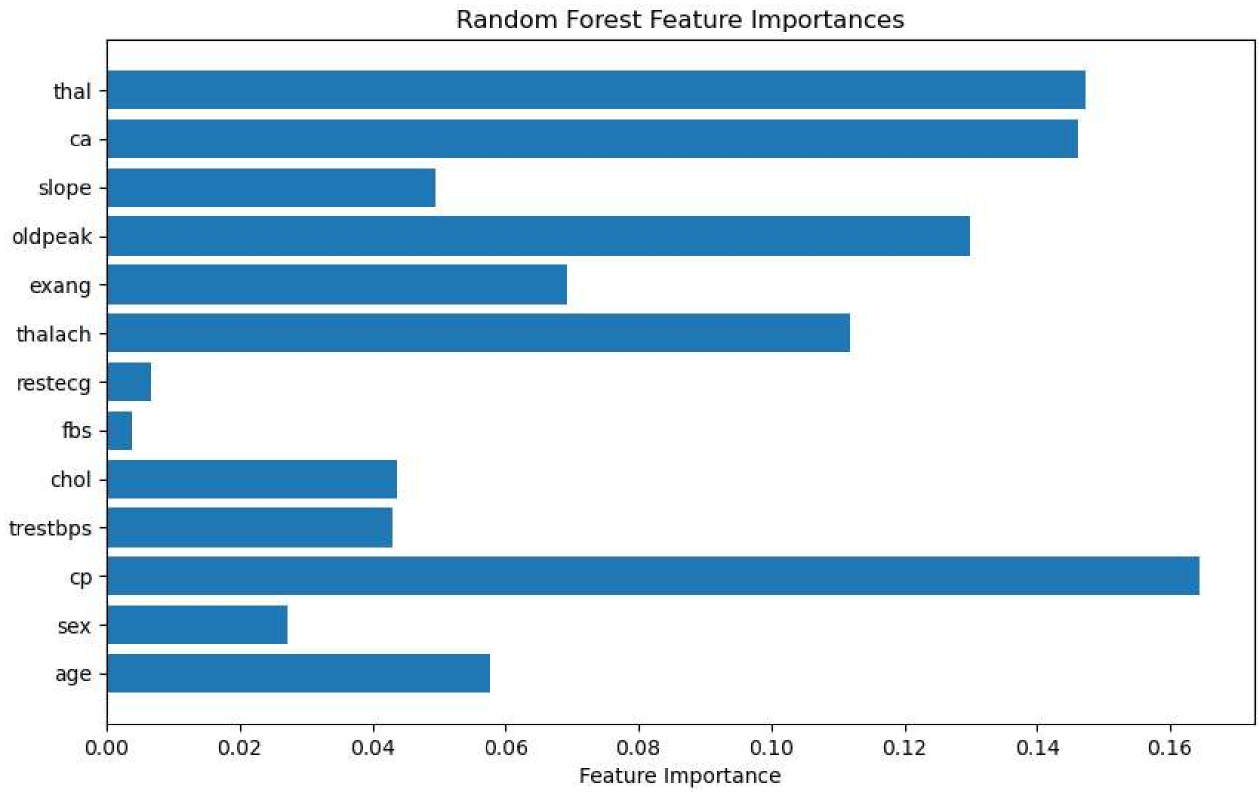
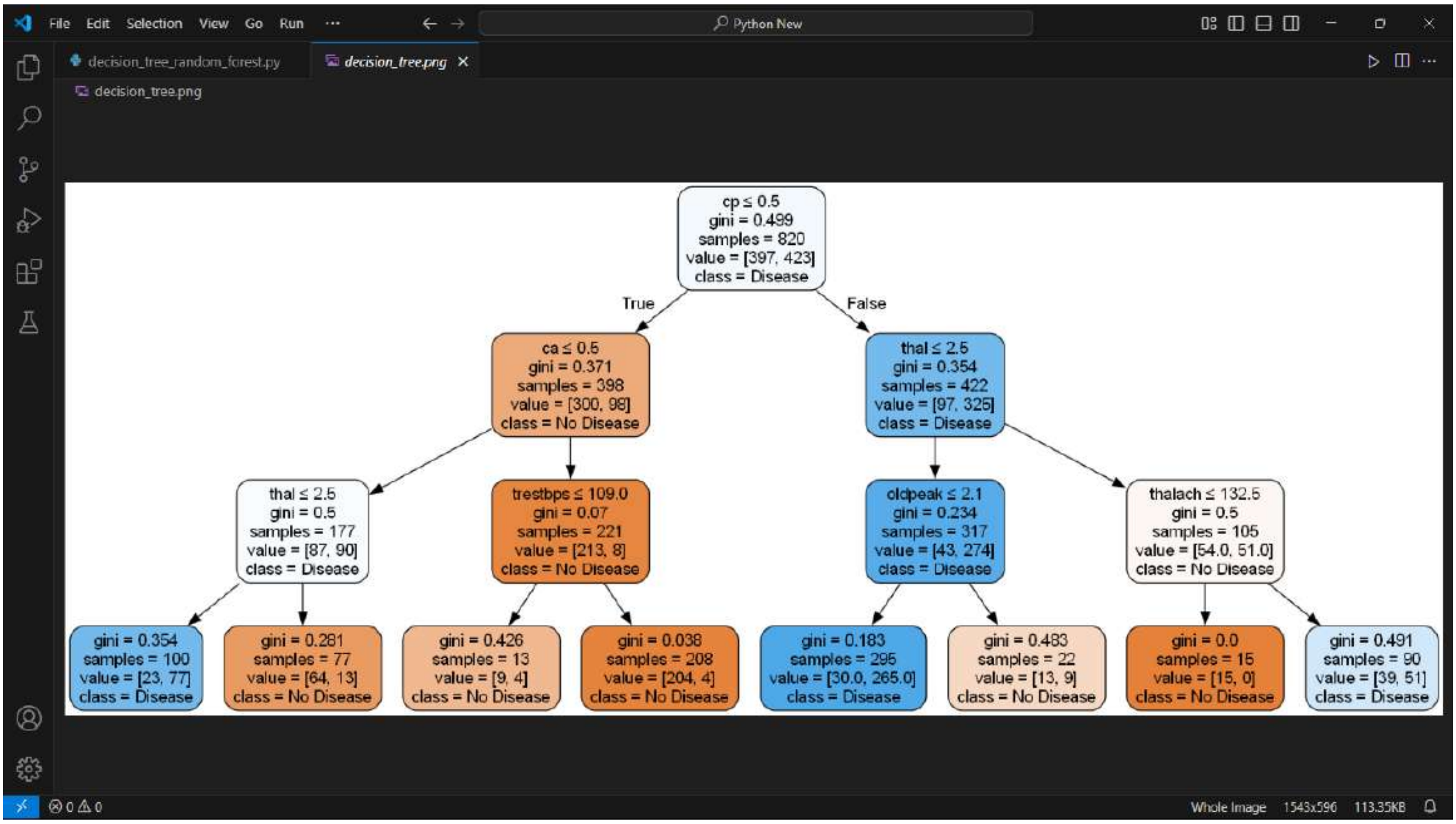
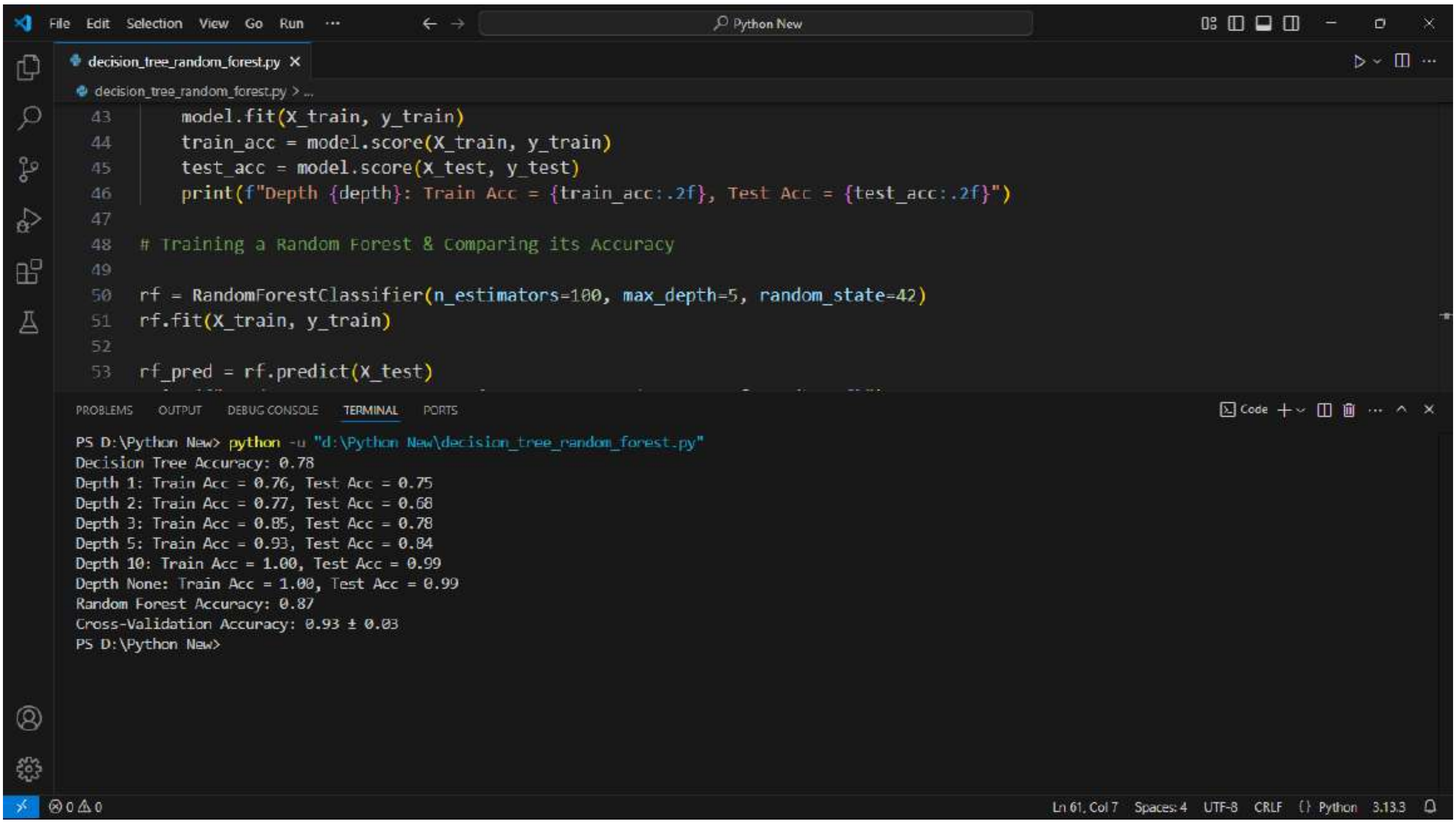

Random Forest Feature Importances
thal
ca
slope
oldpeak
exang
thalach
restecg
fbs
chol
trestbps
cp
sex
age
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Feature Importance


File Edit Selection View Go Run ···
Python New
decision_tree_random_forest.py
decision_tree.png
decision_tree.png
cp ≤ 0.5
gini = 0.499
samples = 820
value = [397, 423]
class = Disease
True
False
ca ≤ 0.5
gini = 0.371
samples = 398
value = [300, 98]
class = No Disease
thal ≤ 2.5
gini = 0.354
samples = 422
value = [97, 325]
class = Disease
thal ≤ 2.5
gini = 0.5
samples = 177
value = [87, 90]
class = Disease
trestbps ≤ 109.0
gini = 0.07
samples = 221
value = [213, 8]
class = No Disease
oldpeak ≤ 2.1
gini = 0.234
samples = 317
value = [43, 274]
class = Disease
thalach ≤ 132.5
gini = 0.5
samples = 105
value = [54.0, 51.0]
class = No Disease
gini = 0.354
samples = 100
value = [23, 77]
class = Disease
gini = 0.281
samples = 77
value = [64, 13]
class = No Disease
gini = 0.426
samples = 13
value = [9, 4]
class = No Disease
gini = 0.038
samples = 208
value = [204, 4]
class = No Disease
gini = 0.183
samples = 296
value = [30.0, 265.0]
class = Disease
gini = 0.483
samples = 22
value = [13, 9]
class = No Disease
gini = 0.0
samples = 15
value = [15, 0]
class = No Disease
gini = 0.491
samples = 90
value = [39, 51]
class = Disease
Whole Image 1543x596 113.35KB


File Edit Selection View Go Run ···
Python New
decision_tree_random_forest.py
decision_tree_random_forest.py > ...
    model.fit(X_train, y_train)
    train_acc = model.score(X_train, y_train)
    test_acc = model.score(X_test, y_test)
    print(f"Depth {depth}: Train Acc = {train_acc:.2f}, Test Acc = {test_acc:.2f}")
# Training a Random Forest & Comparing its Accuracy
rf = RandomForestClassifier(n_estimators=100, max_depth=5, random_state=42)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
PS D:\Python New> python -u "d:\Python New\decision_tree_random_forest.py"
Decision Tree Accuracy: 0.78
Depth 1: Train Acc = 0.76, Test Acc = 0.75
Depth 2: Train Acc = 0.77, Test Acc = 0.68
Depth 3: Train Acc = 0.85, Test Acc = 0.78
Depth 5: Train Acc = 0.93, Test Acc = 0.84
Depth 10: Train Acc = 1.00, Test Acc = 0.99
Depth None: Train Acc = 1.00, Test Acc = 0.99
Random Forest Accuracy: 0.87
Cross-Validation Accuracy: 0.93 ± 0.03
PS D:\Python New>
Ln 61, Col 7 Spaces: 4 UTF-8 CRLF {} Python 3.13.3